RL locomotion steering: hc-height-speed-004

2026-09-08T00:07:11+00:00

Outcome

Recorded run status: confirmation_failed.

The exact height-derived family from hc-running-002 was tested for speed control on a finer signed grid, without refitting or training. Six of twelve primary settings passed validation; the locked choice alpha=-0.02 had the largest eligible slowdown. On thirty fresh confirmation episodes it slowed speed from 16.80155 to 11.44279 m/s (-31.89%; paired effect -5.35876 m/s, 95% CI [-6.98568, -3.99911]) but physically failed in ten episodes (33.33%), so confirmation failed. All 72 validation and six confirmation conditions are retained. Two controls passed confirmation pilot gates, including a shuffled direction with one physical failure within the predeclared allowance; neither control was replicated. No replication or application episodes were collected. The selected strength was not replaced after confirmation.

Validation selection: speed (speed), alpha=-0.02; speed_bias (speed), alpha=-0.025; speed_random0 (speed), alpha=0.04; speed_random1 (speed), alpha=-0.035; speed_random2 (speed), alpha=-0.035; speed_shuffled (speed), alpha=-0.035

A gate pass is a pilot usefulness decision for one condition; it does not establish generalization across training seeds.

Question and method

Can adding a constant vector after the first actor ReLU change locomotion while preserving movement? The policy stays frozen. Vectors are episode-weighted differences of contrasting unsteered activations, normalized to centered activation RMS. The protocol includes random and shuffled-label controls on the same evaluation split when that stage is reached. No behavioral cloning is used.

Interpretation

Delta is treated minus baseline in the behavior's units. Speed/lateral use m/s, height uses m, turning uses rad/s, and effort is mean squared normalized action, not physical energy. Confidence intervals resample paired episodes. Validation selects strengths; confirmation and replication provide fresh evidence when recorded. Useful effects require the target threshold, a confidence interval excluding zero, and locomotion/quality checks. Control effects must be considered before attributing specificity to the extracted direction.

Fitting diagnostics

Inherited fitting diagnostics from runs\hc-running-002; source vector height is evaluated as speed. No new fitting episodes were collected. The source statistics below retain their original labels.

Fitting windows: episodes=64; total windows=576; eligible windows=564; eligible episodes=64; excluded nonfinite=0; excluded unhealthy=12; discarded tail steps=0; warmup=100; window=100; minimum speed fraction=0.5; minimum speed floor=7.327; healthy median speed=14.654; excluded slow windows=0; speed filter reason=project pilot heuristic: exclude finite healthy fitting windows at or below fraction times their median speed.

Centered activation RMS: 8.705. Convention: episode-balanced eligible timestep RMS about episode-balanced mean; full 100-step windows after 100-step warmup.

effort (mean squared action): status=fitted; high episodes=60; low episodes=57; high windows=144; low windows=144; contrast=0.031536; raw norm=0.43919; split-half cosine=0.89514.

height (m): status=fitted; high episodes=61; low episodes=61; high windows=141; low windows=141; contrast=0.044745; raw norm=1.2007; split-half cosine=0.97976.

speed (m/s): status=fitted; high episodes=61; low episodes=59; high windows=141; low windows=141; contrast=2.3066; raw norm=1.021; split-half cosine=0.98237.

Split-half cosine measures extraction agreement, not causal effectiveness. Full bin statistics and vector coordinates are retained in vector_diagnostics.json.

Research decisions

2026-09-07T23:58:41.398747+00:00: Test whether the unchanged height-contrast vector from runs\hc-running-002 provides useful speed control. Recalibrate its complete control family on fresh validation episodes, then freeze selection before confirmation. Evidence: Parent validation suggested a cross-behavior effect. This is an explicitly adaptive exploratory follow-up; it does not identify an internal semantic concept..

Validation

72 recorded intervention conditions.

Vector / behavior	Alpha	Delta	95% CI	Pairs	Useful
speed / speed	-0.04	-7.4797	[-10.357, -4.7672]	10	no
speed / speed	-0.035	-8.0016	[-10.706, -5.418]	10	no
speed / speed	-0.03	-3.6865	[-4.2169, -3.3239]	10	no
speed / speed	-0.025	-5.7659	[-8.1115, -3.6734]	10	no
speed / speed	-0.02	-2.7618	[-2.9754, -2.4958]	10	yes
speed / speed	-0.01	-1.9757	[-2.681, -1.5042]	10	no
speed / speed	0.01	-0.51444	[-0.61165, -0.44272]	10	no
speed / speed	0.02	-1.128	[-1.2014, -1.0632]	10	yes
speed / speed	0.025	-1.3134	[-1.4185, -1.2223]	10	yes
speed / speed	0.03	-1.5581	[-1.6441, -1.4784]	10	yes
speed / speed	0.035	-1.7577	[-1.8372, -1.6885]	10	yes
speed / speed	0.04	-2.0042	[-2.173, -1.8695]	10	yes
speed_random0 / speed	-0.04	-0.39776	[-0.50032, -0.30182]	10	no
speed_random0 / speed	-0.035	-0.33028	[-0.40842, -0.25486]	10	no
speed_random0 / speed	-0.03	-0.23395	[-0.3126, -0.16407]	10	no
speed_random0 / speed	-0.025	-0.12049	[-0.13308, -0.10998]	10	no
speed_random0 / speed	-0.02	-0.10242	[-0.10515, -0.098471]	10	no
speed_random0 / speed	-0.01	-0.096225	[-0.10483, -0.087207]	10	no
speed_random0 / speed	0.01	-0.11471	[-0.18705, -0.05017]	10	no
speed_random0 / speed	0.02	-0.38219	[-0.42353, -0.34287]	10	no
speed_random0 / speed	0.025	-0.5627	[-0.61675, -0.51031]	10	no
speed_random0 / speed	0.03	-0.69011	[-0.74616, -0.63852]	10	no
speed_random0 / speed	0.035	-0.81081	[-0.87936, -0.75068]	10	no
speed_random0 / speed	0.04	-1.2025	[-1.9955, -0.7751]	10	no
speed_random1 / speed	-0.04	-1.4713	[-2.2733, -0.96726]	10	no
speed_random1 / speed	-0.035	-1.6807	[-3.3753, -0.74555]	10	no
speed_random1 / speed	-0.03	-0.36577	[-0.47083, -0.26464]	10	no
speed_random1 / speed	-0.025	-0.222	[-0.32525, -0.14824]	10	no
speed_random1 / speed	-0.02	-0.12163	[-0.14629, -0.093287]	10	no
speed_random1 / speed	-0.01	-0.0431	[-0.057729, -0.031301]	10	no
speed_random1 / speed	0.01	0.0029704	[-0.0024723, 0.0082149]	10	no
speed_random1 / speed	0.02	0.02272	[0.018168, 0.027824]	10	no
speed_random1 / speed	0.025	0.029461	[0.023554, 0.035611]	10	no

Vector / behavior	Alpha	Delta	95% CI	Pairs	Useful
speed_random1 / speed	0.03	0.035312	[0.029835, 0.041088]	10	no
speed_random1 / speed	0.035	0.032951	[0.026911, 0.038998]	10	no
speed_random1 / speed	0.04	0.018667	[0.012031, 0.025362]	10	no
speed_random2 / speed	-0.04	-0.67658	[-1.4594, -0.21129]	10	no
speed_random2 / speed	-0.035	-1.5165	[-3.8325, -0.25186]	10	no
speed_random2 / speed	-0.03	-0.32184	[-0.54758, -0.13681]	10	no
speed_random2 / speed	-0.025	-0.026994	[-0.060267, 0.0019618]	10	no
speed_random2 / speed	-0.02	-0.16029	[-0.22085, -0.1091]	10	no
speed_random2 / speed	-0.01	-0.06245	[-0.10678, -0.029006]	10	no
speed_random2 / speed	0.01	0.0049307	[-0.0069636, 0.015473]	10	no
speed_random2 / speed	0.02	-0.22658	[-0.57698, -0.03101]	10	no
speed_random2 / speed	0.025	-0.078226	[-0.14479, -0.023982]	10	no
speed_random2 / speed	0.03	-0.021559	[-0.029636, -0.01371]	10	no
speed_random2 / speed	0.035	-0.037576	[-0.065754, -0.012835]	10	no
speed_random2 / speed	0.04	-0.017796	[-0.030001, -0.0087582]	10	no
speed_shuffled / speed	-0.04	-2.5354	[-3.0131, -2.1877]	10	no
speed_shuffled / speed	-0.035	-2.1063	[-2.3152, -1.9049]	10	yes
speed_shuffled / speed	-0.03	-1.6254	[-1.8032, -1.4716]	10	yes
speed_shuffled / speed	-0.025	-1.2259	[-1.3098, -1.1531]	10	yes
speed_shuffled / speed	-0.02	-1.0313	[-1.1614, -0.94414]	10	yes
speed_shuffled / speed	-0.01	-0.46926	[-0.54605, -0.40082]	10	no
speed_shuffled / speed	0.01	-0.60532	[-0.77018, -0.43862]	10	no
speed_shuffled / speed	0.02	-2.0407	[-3.6859, -1.0647]	10	no
speed_shuffled / speed	0.025	-4.1548	[-7.4316, -1.7853]	10	no
speed_shuffled / speed	0.03	-9.814	[-12.809, -6.4459]	10	no
speed_shuffled / speed	0.035	-7.9126	[-11.153, -4.9343]	10	no
speed_shuffled / speed	0.04	-6.5451	[-9.7185, -3.4643]	10	no
speed_bias / speed	-0.04	0.048573	[-0.021633, 0.098707]	10	no
speed_bias / speed	-0.035	0.067946	[0.044758, 0.08764]	10	no
speed_bias / speed	-0.03	-0.3816	[-0.52577, -0.23823]	10	no
speed_bias / speed	-0.025	-0.59631	[-1.5459, -0.071972]	10	no
speed_bias / speed	-0.02	0.010885	[-0.036184, 0.044599]	10	no
speed_bias / speed	-0.01	0.014225	[-0.024591, 0.037364]	10	no
speed_bias / speed	0.01	-0.074037	[-0.10232, -0.050679]	10	no
speed_bias / speed	0.02	-0.11299	[-0.16114, -0.081494]	10	no
speed_bias / speed	0.025	-0.13998	[-0.15784, -0.12343]	10	no
speed_bias / speed	0.03	-0.1828	[-0.20696, -0.16462]	10	no
speed_bias / speed	0.035	-0.22313	[-0.2941, -0.16987]	10	no
speed_bias / speed	0.04	-0.26185	[-0.32872, -0.20332]	10	no

Confirmation

6 recorded intervention conditions.

Vector / behavior	Alpha	Delta	95% CI	Pairs	Useful
speed / speed	-0.02	-5.3588	[-6.9857, -3.9991]	30	no
speed_bias / speed	-0.025	-0.11285	[-0.16153, -0.063808]	30	no
speed_random0 / speed	0.04	-0.81328	[-0.87674, -0.76671]	30	no
speed_random1 / speed	-0.035	-0.85519	[-0.96866, -0.76348]	30	yes
speed_random2 / speed	-0.035	-0.55205	[-0.97725, -0.31906]	30	no
speed_shuffled / speed	-0.035	-2.3381	[-3.1976, -1.8624]	30	yes

Unmeasured stages

No recorded measurements: replication. No causal steering outcome is inferred for these stages.

Reproduction and provenance

Created: 2026-09-07T23:58:41.364452+00:00; code commit: 2377452.

Policy: farama-minari/HalfCheetah-v5-TQC-medium; algorithm: TQC; environment: HalfCheetah-v5.

Checkpoint revision: b4ce04da6f246f06ae4c4258b8ab39624e6600b4. SHA256: 8231919095d5a35a2b799e865e968feca535d5af3f40a4b092947c70a976b7e1.

Layer: actor.latent_pi.1; hidden dimension: 256; observation/action shapes: [17] / [6].

Seed partitions below are registered assignments, not completion counts.

Diagnostic: 16 seeds; range 100000-100015.

Fit: 64 seeds; range 101000-101063.

Validation: 10 seeds; range 310000-310009.

Confirmation: 30 seeds; range 320000-320029.

Replication: 30 seeds; range 330000-330029.

Configuration: torch_threads=1; rollout_workers=4; max_steps=1000; warmup=100; diagnostic_episodes=16; fit_episodes=64; validation_episodes=10; confirmation_episodes=30; replication_episodes=30; seed_offset=3e+05; fit_min_speed_fraction=0.5; strength_grid=-0.04,-0.035,-0.03,-0.025,-0.02,-0.01,0.01,0.02,0.025,0.03,0.035,0.04.

Software: numpy 1.26.4; torch 2.5.1+cpu; torchvision 0.20.1+cpu; stable-baselines3 2.4.1; sb3-contrib 2.4.0; gymnasium 1.0.0; mujoco 3.1.5; baukit 0.0.1.

Full exact seed lists, checkpoint metadata and the original specification remain in adjacent manifest.json. Full measured analysis is in results.json; extraction details are in vector_diagnostics.json. The report does not change these artifacts or rerun inference.

Sources

CAA method: https://arxiv.org/abs/2312.06681

Baukit hooks: https://github.com/davidbau/baukit/blob/9d51abd51ebf29769aecc38c4cbef459b731a36e/baukit/nethook.py

Environment and measurement references: docs/sources.md in the repository.

Reporting: https://docs.reportlab.com/reportlab/userguide/ and https://matplotlib.org/stable/users/index.html

Validation strength response: speed